TUTORIAL 3 (EXERCISE 3)

QUESTION 1

Table 6 showing the number of hospitalizations for cardiovascular disease, by age group, for one year. One of these case records is selected at random.

Table 6

Age group (year)	Number
0-19	60
20-39	150
40-49	300
50-59	580
60-69	678
70-79	1288
80 and over	1378

Find the probability that the patient was

a) In the age of 50s [2 Marks]
b) Less than 50 years old. [2 Marks]
c) Between 40 and 69 years old, inclusive [3 Marks]
d) 70 years old or older. [3 Marks]

(a) $P(E) = \frac{a}{b}$

$b = 4434$

$a = \frac{580}{4434} = 0.13\#$

(b) $P(X < 50) = \frac{510}{4434} = 0.115\#$

(c) $P(40 \leq X \leq 69) = \frac{1558}{4434} = 0.35\#$

(d) $P(X \geq 70) = \frac{2666}{4434} = 0.60\#$

QUESTION 2

In a Japanese restaurant, an average of 3 out of every 5 customers ask for plain water with their meal. A random sample of 10 customers is selected. Let X equals to the number of customers that ask for plain water. Assume each trial is independent.

a) Find the probability that exactly 6 ask for plain water with their meal. [3 Marks]
b) Find the probability that less than 9 customers ask for plain water with their meal? [3 Marks]
c) A second random sample of 30 customers is selected. What is the probability that the first who to ask for plain water is the fourth customer. [4 Marks]

$p = 3/5 = 0.6$; $q = 1 - p = 1 - 0.6 = 0.4$

$X \sim B(n=10, p=0.6)$

$$P(X=x) = \binom{n}{x} p^x q^{n-x}$$

(a) $P(X=6) = \binom{10}{6} (0.6)^6 (0.4)^{10-6=4}$

$\binom{10}{6} = \frac{10!}{6!(10-6)!} = 210$ permutation!

$= (210)(0.6)^6(0.4)^4 = 0.25$ *

(b) $P(X<9) = 1 - P(X=9) - P(X=10)$

$P(X=9) = \binom{10}{9} (0.6)^9 (0.4)^1 =$ a(0.04)

$P(X=10) = \binom{10}{10} (0.6)^{10} (0.4)^0 =$ b(0.006)

$P(X<9) = 1 - a - b = 1 - 0.04 - 0.006$

$= 0.95$ *

(c) $p = (0.4)^3(0.6)$ = ____ *

$q^{x-1} \cdot p$

* p(first success at x) $= (1-p)^{x-1} \cdot p$

QUESTION 3

Smart phones has been one of the most important part in our daily life. The length of working time for a smart phone before it needs to be charged, X is normally distributed with a mean of 20 hours and a standard deviation of 2 hours.

a) What is the probability that a phone work for at least a day without charging? [3 Marks]
b) Find the value of X given $P(Z) = 0.12$. [3 Marks]
c) You are going to attend a 2-hour meeting right now and your phone has been used for 25 hours since the last charge. What is the probability that your phone will still work throughout the meeting without charging? [4 Marks]

$X \sim N(\mu = 20, \sigma = 2)$

$Z = \frac{X - \mu}{\sigma}$

P(0 < Z < 2)

(a) $P(X \geq 24)$

$Z = \frac{24 - 20}{2} = \frac{4}{2} = 2$

$P(X \geq 24) = P(Z \geq 2)$

$P(Z \leq 2) = 0.9772$

$P(Z \geq 2) = 1 - 0.9772$

$= 0.0228$ #

$\therefore P(X \geq 24) = 0.0228$ #